



Will the Free Market Respond to Global Birth Rate Decline?

A Synthesis of the Evidence on Global Fertility Decline

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Table of Contents

Table of Contents	2
1 Executive Summary	3
2 Introduction: The Global Fertility Crisis and Contemporary Discourse	4
3 Demographic Shifts	4
3.1 The Global Fertility Transition	4
4 Who Is Childless and Why: Measurement Challenges and the Invisible Male.....	6
4.1 The Measurement Problem: Voluntary vs. Involuntary Childlessness	6
4.2 The Invisible Male	8
5 Economics: Necessary But Not Sufficient	9
5.1 What Interventions Tell Us	9
5.2 The Structural Transformation of Child-Rearing Costs.....	9
5.3 The Motherhood Penalty and the Fatherhood Premium.....	10
5.4 The Education-Fertility U-Curve and the Middle-Class Squeeze.....	11
5.5 Housing Access and Marriage Rates: Two Streams of Causal Evidence	12
5.6 The Sandwich Generation: The Hidden Double Burden	12
6 Biological Factors: Is This a Medical Crisis?	13
6.1 Infertility, Subfecundity, and Social Childlessness	14
7 Cultural Factors: Partnership Decline, Future Anxiety, and Neurochemical Displacement	15
7.1 Partnership Collapse.....	15
7.2 Secularisation and Individualisation.....	16
7.3 Concern About the Future of the World.....	17
7.4 The Neurochemical Displacement Hypothesis	17
8 Market Responses: What the Free Market Has Tried	18
8.1 Biological Decline: The Fertility Economy	18
8.2 Partnership Failure: Dating Apps and Their Perverse Effects.....	18
8.3 Neurochemical Displacement: AI Companions.....	19
8.4 Economic Barriers: Pronatalist Transfers and Their Limits	19
8.5 The Furbaby Economy: Pets as a Partial Substitute	20
9 Is Declining Fertility Actually a Crisis?.....	20
10 Conclusion.....	21
11 References.....	23

1 Executive Summary

The global total fertility rate (TFR), the average number of children born per woman, has fallen from 4.8 in 1950 to 2.2 in 2023. Two thirds of humanity now lives in countries where the birth rate has dropped below the replacement level of 2.1 children per woman. Forty-six countries have entered what demographers call lowest-low fertility, with rates below 1.5, including South Korea at 0.75 and Taiwan at 0.89. This is not a regional story. It is a global one, and it is accelerating.

The data reveals something counterintuitive about how this decline is happening. Stephen Shaw's (2025) microdemographic framework, published in *Scientific Reports*, decomposes TFR into two components: the Total Maternal Rate (TMR), the share of women who become mothers at all, and Children Per Mother (CPM), the average family size among those who do. The TMR has fallen from approximately 85% to 72% across high-income countries since the early 1970s. The CPM has remained stable at 2.0 to 2.3 (Shaw 2025; Zeman, Beaujouan, Brzozowska, and Sobotka 2018). Fertility decline is concentrated on the extensive margin: fewer adults entering parenthood, not smaller families among those who do. Japan's case makes this concrete. Married couples there average 1.9 children, close to replacement, yet the national TFR is 1.2. The arithmetic gap is explained entirely by the collapse of marriage formation, not by any change in family size preferences.

The biological picture is real but limited in its explanatory power. Sperm concentration has declined 52.4% since 1973. Clinical infertility rates sit at 11 to 15%. These are genuine medical trends that deserve attention. But clinical infertility rates have not risen proportionally to childlessness rates, which have nearly doubled in some cohorts. The average number of children per man aged 15 to 49 in the United States is approximately 0.9. Most childlessness is social, not medical.

Economic factors are necessary but insufficient explanations. Housing unaffordability concentrated in the early to mid-twenties window, the child cost explosion, the 42% maternal earnings penalty in the United States, the invisible double burden of sandwich caregiving, and the middle-class squeeze all operate as real constraints. Hungary's pronatalist programme allocating approximately 6% of GDP has produced TFR gains of only 0.1 to 0.25. If fertility decline were purely an economics problem, larger transfers would work better than they demonstrably do.

Cultural and neurochemical factors operate independently of economic conditions. Partnership collapse, collective pessimism about the future, and the progressive recalibration of the dopamine reward system by digital technology each suppress fertility through mechanisms that no financial transfer addresses. The gender dimension of this runs through the data: male childlessness consistently exceeds female childlessness across every country where data exists, yet it remains almost entirely absent from policy discourse.

Counterintuitively, women with doctoral-level education show higher fertility than those with bachelor's degrees. The fertility nadir falls in the middle-income, college-educated stratum most squeezed between childcare costs and opportunity costs. The fertility crisis is

not a failure of female ambition. It is a structural failure of the economic arrangements governing family formation in middle-income households.

No single policy instrument has reversed demographic decline. The analysis closes by proposing that the most important upstream intervention may not be financial at all. It is restoring the conditions under which young people feel purposeful, financially secure, biologically healthy, and capable of forming genuine romantic partnerships without technological mediation.

2 Introduction: The Global Fertility Crisis and Contemporary Discourse

In early 2025, the global fertility crisis moved from demographic journals to popular culture and geopolitical strategy. The American legislative landscape was reshaped by the One Big Beautiful Bill Act, which codified Trump Accounts: a pro-family initiative providing \$1,000 government contributions for every newborn. Simultaneously, Russian billionaire Pavel Durov announced he has fathered over 100 children via sperm donation and is financing full in-vitro fertilisation protocols for women wishing to use his genetic material. These initiatives follow years of advocacy from Elon Musk characterising population collapse as an existential threat exceeding climate change. The responses are varied, occasionally bizarre, and share one thing in common: they are all downstream of a problem nobody has yet managed to reverse.

In early 2026, a podcast between Steven Bartlett and Chris Williamson attributed declining birth rates primarily to women's workforce participation, provoking widespread backlash. The controversy illustrated how readily fertility discourse collapses into gender blame when untethered from demographic evidence.

What the data actually shows is more structurally interesting, and more difficult, than any single-cause narrative. Parenthood has shifted from a cornerstone of adult life to a capstone: something that requires the prior achievement of financial security, stable partnership, adequate housing, and sufficient confidence in the future before it can even be contemplated. For a growing share of young adults in advanced economies, those conditions are not being met. This paper applies a multi-mechanism analytical framework to understand why. The analysis is organised around the extensive margin thesis, that observed decline is concentrated in the share of adults entering parenthood at all rather than in the family sizes of those who do, and examines the economic, biological, and cultural forces that operate on that margin. The evidence assembled here is clear: no single intervention has successfully reversed declining birth rates, and the problem sits further upstream than most of the responses currently being applied.

3 Demographic Shifts

3.1 The Global Fertility Transition

The global fertility transition represents one of the most profound demographic shifts in human history. Total fertility rates declined from 4.8 children per woman in 1950 to 2.2 in

2023, with two thirds of humanity now residing in countries with below-replacement fertility (Bhattacharjee et al. 2024; World Bank n.d.). Forty-six countries have entered lowest-low fertility territory with rates below 1.5 children per woman, including South Korea at 0.75, Taiwan at 0.89, and Hong Kong at 1.24. Eight of the ten lowest-fertility countries cluster in East Asia or Southern and Eastern Europe, suggesting regional institutional and cultural patterns beyond purely market-driven forces.

See Appendix Figure A1 for a choropleth map of global fertility rates by country (2023), illustrating the full geographic extent of below-replacement fertility.

This demographic transition creates substantial economic implications. Advanced economies face dependency ratios declining from 3.2 working-age individuals per senior today to 2.0 by 2050 and 1.4 by 2100, while China's ratio shifts from 4.8 to 1.2 over the same period (United Nations 2024). These structural shifts generate fiscal pressures on pension and healthcare systems designed when post-retirement dependency periods were minimal. When UK pension systems were established mid-century with payments beginning at age 65, male life expectancy was 66. Today it reaches 79, creating 15 or more years of dependency funded by shrinking workforces (ONS n.d.).

See Appendix Figures A4 and A5 for the world population by age group projected to 2100, and the US population pyramid transformation from 2000 to 2020, which illustrate the structural inversion of age distributions underlying these dependency projections.

Regional patterns reveal a three-wave structure. First-wave regions, advanced economies in North America, Western Europe, and East Asia, crossed below replacement fertility in the 1970s to 1990s and now average 1.5 to 1.7 children per woman. Second-wave regions including Latin America and much of Asia fell below replacement in the 2000s to 2010s. Third-wave regions, primarily Sub-Saharan Africa, maintain fertility above 4.0 but are declining rapidly, projected to reach 2.0 by 2100 (United Nations 2024).

Standard demographic transition theory posits that fertility declines represent rational adjustment to falling mortality. Since 1950, global average life expectancy has increased from approximately 46 to 73 years. Global infant mortality has declined from approximately 19.8% in 1950 to 3.7% in 2023 (UN IGME n.d.). As child survival rates improved, the insurance strategy of having large families to ensure surviving offspring became less necessary. However, this longevity dividend has created secondary challenges. The expansion of post-retirement dependency periods strains public welfare systems, with developed nations facing widespread pension underfunding.

The global population currently exists in a state of demographic momentum that masks the severity of fertility decline. Although fertility rates have fallen below replacement level in nations housing two thirds of humanity, absolute global population continues growing (United Nations 2024). This paradox reflects the structural composition of age pyramids: cohorts currently at the oldest ages are numerically smaller than cohorts of childbearing age. But this buffer is temporary. The oldest Baby Boomers, born in 1946, are now 80, with the entire generation entering the dependency window over the next two decades. Current projections indicate that natality and mortality curves will intersect in the latter half of the 21st century, at which point total human population will enter sustained, structural decline

(United Nations 2024). The remainder of this paper examines the economic, biological, and cultural forces driving this transition in the countries that have experienced it longest, and what, if anything, can be done about it.

See Appendix Figure A2 for the global birth rate and death rate trajectories from 1950 to 2100, showing the projected crossover point at which natural population decline becomes structural.

4 Who Is Childless and Why: Measurement Challenges and the Invisible Male

4.1 The Measurement Problem: Voluntary vs. Involuntary Childlessness

Looking more closely at the data from the first countries to experience sustained fertility decline, an interesting picture emerges. Average family size has stayed roughly the same. What has changed is the number of women becoming mothers at all. Stephen Shaw's (2025) microdemographic framework, published in Scientific Reports, formalises this decomposition: TFR equals the Total Maternal Rate multiplied by Children Per Mother. The TMR has fallen from approximately 85% to 72% across high-income countries since the early 1970s, while CPM has held steady at 2.0 to 2.3 (Shaw 2025; Zeman, Beaujouan, Brzozowska, and Sobotka 2018). Fertility decline is not about people having smaller families. It is about more people not starting families at all. The question of whether this shift is voluntary or involuntary, and what is driving it, is examined in the cultural section of this paper. What the data here reveals first is how difficult that question is to answer.

Distinguishing voluntary from involuntary childlessness presents profound methodological challenges. Survey-based measurements typically rely on retrospective self-reports, in which respondents categorise their childlessness as chosen or unchosen. But retrospective reports are subject to rationalisation. Individuals who were unable to achieve parenthood due to structural barriers, lack of a partner, insufficient income, housing insecurity, may over time reconstruct their experience as a preference, particularly where voluntary childlessness carries lower social stigma than involuntary failure. This rationalisation of unavoidable outcomes represents a fundamental validity threat to survey-based decompositions of childlessness by volition.

An apparent contradiction in available survey data illustrates the measurement problem. Approximately 10% of women in long-run surveys report zero desired children, a figure that has been relatively stable and suggests a small but consistent minority for whom voluntary childlessness represents a genuine, stable preference formed independently of economic constraints. Yet the 2024 Pew Research Center survey of childless and child-free Americans found that 57% of respondents cited personal desire as a major reason for their childlessness, a figure far exceeding the 10% who prospectively report zero desired children. This contradiction is resolved not by concluding that one source is incorrect but by recognising that personal desire, as retrospectively reported, is not equivalent to always wanting to be childfree. It reflects preferences that may have formed adaptively in response to structural constraints, with the constraint becoming internalised as preference over time.

The scale of the phenomenon lends urgency to this analytical question. Research by Kenneth Johnson of the University of New Hampshire estimates that as of 2024, there are approximately 5.7 million more childless women in the United States than would be predicted by historical fertility patterns (Johnson 2024). The Census Bureau's Current Population Survey documents that childlessness among women aged 20 to 24 has risen from 75% to 85% between 2012 and 2024, and among women aged 30 to 34 from 29% to 40% over the same period. A crucial finding moderates the interpretation of these rising rates: childlessness among women aged 45 to 50, those effectively past reproductive age, has decreased from 16.7% to 14.9% over the same period. This decline in completed childlessness at the end of the reproductive window, juxtaposed against rising childlessness at younger ages, is consistent with a tempo effect. Childbearing is being delayed rather than abandoned for many women, even as the aggregate stock of childlessness at any given point in time appears to rise.

See Appendix Figure A3 for cohort-level data on age of mothers at childbirth across Sweden, Spain, the United States, and Japan, which visualises this tempo shift directly: successive cohorts are older at each birth order, with younger cohorts' curves barely registering.

The Pew (2024) decomposition of reasons for childlessness reveals the composite character of the phenomenon. Among respondents identifying as childless or child-free, 57% cited personal desire as a major factor, 44% cited medical reasons, 43% cited financial constraints, 38% reported the absence of a suitable partner, and 32% cited concerns about the state of the world or climate change. These categories are not mutually exclusive. An individual may simultaneously hold a genuine preference for childlessness, face financial barriers that reinforce that preference, and lack a partner with whom parenthood would be contemplated. The co-occurrence of multiple factors suggests that the binary voluntary or involuntary distinction oversimplifies a reality in which preferences, circumstances, and structural constraints interact dynamically. Demographer Tomas Sobotka's concept of preference endogeneity captures this dynamic: preferences for childlessness may be partially constituted by the experience of structural constraints on parenthood, meaning that observed voluntary childlessness partly reflects circumstances first encountered as involuntary.

See Appendix Figure A6 for a 14-country UNFPA survey (2024) on the most common reasons cited for having fewer children than desired, which documents the composite, cross-national character of this phenomenon.

Over a 17-year observation window, Johnson (2024) estimates a cumulative shortfall of approximately 11.8 million births relative to historical trend projections in the United States alone. Whether this shortfall ultimately reflects permanent quantum decline, recoverable tempo delay, or an evolving mixture of both depends critically on whether the structural conditions that have made partnership formation and childbearing economically difficult for younger cohorts are resolved, remain stable, or intensify.

4.2 The Invisible Male

One of the most consequential blind spots in contemporary fertility research is the systematic undercounting and analytical neglect of male childlessness. The way demographic data is collected reflects a basic practical reality: birth records are organised around the mother, who is always present and identifiable at the point of birth. The father may be absent, unknown, or unrecorded. This has produced a body of literature in which female childlessness is extensively documented, theorised, and politically debated, while male childlessness remains what the sociologist Robin Hadley (2017) termed an invisibility. Men who remain childless are largely absent from demographic registries, policy frameworks, and popular discourse. The consequence is not merely a research gap but a systematic distortion of how societies understand, explain, and respond to fertility decline.

Cross-national survey data, where available, reveal that male childlessness is not only common but consistently exceeds female childlessness by a substantial margin. Across European countries, approximately 25% of men remain childless compared to approximately 20% of women (Miettinen, Rotkirch, and Szalma 2015). In the United States, data from the National Survey of Family Growth for individuals aged 40 to 49 indicate that 76.5% of men have fathered at least one biological child, compared to 84.3% of women who have given birth: an 8 percentage point gap in lifetime parenthood rates (National Center for Health Statistics 2021). In Norway, where exceptional administrative data permit linkage of fertility outcomes to lifetime income records, the socioeconomic stratification of male childlessness is stark. Among men in the lowest-earning 5%, childlessness reaches 72%, compared to approximately 11% among the highest earners, a gap of 61 percentage points (Bratsberg, Kotsadam, and Walther 2021; IZA Discussion Paper No. 14506). This income gradient in male childlessness has widened by approximately 20 percentage points over a 30-year observation period, tracking the rise in income inequality.

Kravdal's (2021) analysis of Norwegian cohort data documents a widening gender gap in childlessness from 5.8 percentage points to 10.2 percentage points across successive birth cohorts, indicating that men are falling further behind women in achieving parenthood over time. Bratsberg et al. (2021) further demonstrate that job loss attributable to firm bankruptcy reduces male fertility for up to 15 years following the displacement event. Financial instability during prime reproductive years generates permanent demographic consequences, not merely temporary delays. This evidence supports the concept of social infertility: a condition in which structurally induced barriers to stable employment, partnership formation, and household formation prevent reproduction that would otherwise occur, independent of any biological incapacity.

The assumption that childless men are predominantly voluntary non-parents is contradicted by attitudinal survey evidence. Hadley and Hanley (2010), in their research on involuntarily childless men, found that over 90% of childless men reported having desired children during their reproductive years. This figure challenges the cultural presumption that men who do not father children are primarily choosing not to do so. A large majority of childless men appear to experience childlessness as a condition imposed by circumstance rather than as an expression of preference.

5 Economics: Necessary But Not Sufficient

5.1 What Interventions Tell Us

The most direct test of whether fertility decline is primarily an economics problem is the evidence from countries that have tried to reverse it through economic transfers. Hungary's pronatalist programme, one of the most ambitious in the world, allocates approximately 6% of GDP through tax exemptions, subsidised loans, and direct transfers. The observed TFR gain is 0.1 to 0.25 children per woman (Szabó 2024), a modest effect that demographic analysis suggests reflects primarily tempo changes rather than quantum increases in completed family size (Sobotka and Beaujouan 2018). The OBBBA's \$1,000 Trump Account represents less than one week of US infant childcare costs. Across OECD countries, cash transfers, tax exemptions, and paid leave schemes consistently produce measurable but modest fertility responses.

The inference from this evidence is analytically important. Economic support is necessary and modestly effective, but no transfer amount has reversed decline. If fertility decline were purely an economics problem, larger transfers would work better than they demonstrably do. The modest effect sizes of even very generous interventions suggest additional mechanisms are operating. Economics are necessary but not sufficient. The subsequent subsections examine what, specifically within economics, operates as a driver, before the cultural section examines what economics cannot explain at all.

5.2 The Structural Transformation of Child-Rearing Costs

The economic theory of fertility, originating with Gary Becker's (1960) foundational work and substantially developed by John Caldwell's (1976) intergenerational wealth flows hypothesis, predicts fertility decline when children transition from economic assets to economic liabilities. Caldwell's framework identifies the pivotal role of mass education in reversing intergenerational wealth flows. In pre-modern agrarian economies, children contributed economically from early ages through labour, serving simultaneously as old-age insurance and wealth-accumulation vehicles for parents. The spread of compulsory education, by removing children from productive labour for extended periods while substantially increasing parental investment requirements, inverted this calculus. Combined with the decline of multigenerational household support structures and the rise of state-provided old-age pensions that partially substituted for offspring as retirement insurance, the economic incentive structure governing fertility was systematically transformed across the twentieth century.

Contemporary child-rearing cost structures in high-income countries reflect this transformation in magnified form. Direct costs of raising a child to adulthood in the United States are estimated at approximately \$300,000 to \$350,000 in 2024 dollars, excluding higher education costs (USDA 2023). Childcare costs now consume 18 to 22% of household income in the United States and United Kingdom for families with children under five, compared to 6 to 7% in higher-fertility countries with publicly subsidised childcare systems such as France and the Nordic countries. These cost structures create direct economic

incentives against having additional children, particularly for middle-income households where childcare costs may approach or exceed one parent's net income.

Garey Ramey and Valerie Ramey's (2010) analysis of the Rug Rat Race documents a secular increase in the time intensity of parenting among college-educated Americans. Between 1975 and 2007, college-educated American parents increased time spent on childcare activities by approximately nine hours per week, more than doubling their childcare time investment. The Rameys argue that intensifying competition for university placement drove escalating parental time investment as a strategy for securing children's educational attainment. This concerted cultivation of children (Lareau 2003) raises the effective cost of each child, since each additional child competes for parental time of rapidly escalating per-child required quantity, while simultaneously reducing the marginal time available per child in a multi-child household.

When parents bear the full cost of raising children but the wider economy benefits from having a future workforce and tax base, there is a fundamental mismatch between who pays and who gains. Parents fund the next generation of workers whose taxes will support everyone's pensions and healthcare, yet receive no direct return on that investment. This gap between the private cost of having children and the public benefit of more taxpayers creates a strong case for public subsidy of childbearing, not just as a welfare measure but as a correction of a genuine market failure.

5.3 The Motherhood Penalty and the Fatherhood Premium

Labour market analysis consistently documents a substantial penalty on women's earnings associated with childbearing, contrasted with a frequently documented premium on men's earnings associated with fatherhood. Kleven, Landais, and Søgaaard's (2019) landmark study using Danish administrative data pioneered the child penalty methodology, tracking earnings trajectories of individuals before and after childbearing while controlling for individual and partner characteristics. Their analysis finds that childbirth generates approximately a 22% long-run earnings penalty for Danish women, one of the most egalitarian labour market contexts in the world. In the United States, where workplace and childcare policies are less supportive of maternal employment continuity, the estimated child penalty on women's earnings reaches approximately 42% (Kleven et al. 2019). These penalties reflect a combination of reduced hours, career interruptions, occupational downgrading, and reduced progression within firms.

Simultaneously, research consistently documents a fatherhood premium: men's earnings increase following the birth of a child, a pattern documented across the United States, the United Kingdom, Germany, and Scandinavian countries (Lundberg and Rose 2000). The asymmetry between the child penalty on mothers and the child premium on fathers represents a substantial economic structure that shapes individual incentives. For heterosexual couples contemplating childbearing, the economically rational decision, absent external intervention, is for the higher-earning partner to maintain labour force attachment while the lower-earning partner assumes primary caregiving responsibility. For middle-income households, the magnitude of the maternal child penalty, a 42% earnings

reduction on an annual salary of \$75,000 representing a \$31,500 annual loss, creates very strong economic incentives against, or at least toward substantial delay of, first childbirth.

Doepke and Kindermann's (2022) analysis of the intra-household bargaining dynamics of fertility decisions provides complementary evidence. Their work demonstrates that in low-fertility countries, women often oppose additional children beyond the first or second because they anticipate bearing a disproportionate share of the associated childcare burden, even in partnerships with equal gender attitudes. Only 40% of married families with young children can manage on a single income in the United States in 2024, compared to approximately 70% in the 1970s (Institute for Family Studies 2024), making the dual-income household the economic baseline against which family formation feasibility is assessed.

5.4 The Education-Fertility U-Curve and the Middle-Class Squeeze

Standard demographic theory and popular discourse alike assume a monotone negative relationship between women's educational attainment and fertility: higher education delays marriage, increases opportunity costs of childbearing, and tends to be associated with lower completed family size. While this relationship holds broadly in aggregate cross-national comparisons and for early-industrial fertility transitions, research on contemporary high-income populations reveals a more nuanced U-shaped relationship between educational attainment and fertility that challenges simple linear narratives.

Hazan and Zoabi's (2015) theoretical framework and supporting empirical analysis demonstrate that highly educated women, specifically those with doctoral-level or equivalent qualifications, demonstrate higher fertility than women with bachelor's degrees, despite having the highest opportunity costs of childbearing. The mechanism proposed is marketisation: women with sufficiently high earnings can afford to purchase high-quality childcare services at market rates, enabling labour force continuity that mitigates the career penalty of motherhood while maintaining family formation. A doctoral holder earning \$150,000 to \$200,000 annually can absorb nanny or premium childcare costs of \$30,000 to \$40,000 per year as a large but manageable expense. A college-educated woman earning \$60,000 to \$80,000 cannot. The fertility rate among women without a high school diploma stands at approximately 2.3 to 2.5 children; among high school graduates, 2.0 to 2.1; among those with some college but no degree, 1.8 to 1.9; among those with a bachelor's degree, approximately 1.7 to 1.8, the nadir of the distribution; and among those with advanced or doctoral degrees, approximately 1.9 to 2.1, recovering toward the higher end.

This pattern is consistent with the middle-class squeeze thesis: women earning between approximately \$50,000 and \$80,000 annually face the worst trade-offs. They earn too much for the opportunity cost of workforce exit to be negligible, but too little to absorb formal childcare costs without household financial strain reaching levels that depress desired family size. The middle-class squeeze concentrates fertility challenges among precisely the group most associated with stable, long-term-oriented family formation, and the most numerically significant educational stratum in developed economies.

5.5 Housing Access and Marriage Rates: Two Streams of Causal Evidence

The link between housing access and fertility has moved from theoretical plausibility to empirically demonstrated causal significance. Van Doornik, Fazio, Ramadorai, and Skrastins (2024) studied what happened to fertility outcomes among applicants for subsidised housing in Brazil, where units were allocated by lottery. Because the lottery was random, it created a clean natural experiment: the only difference between winners and losers was whether they got the housing, not anything about their intentions or circumstances. Their analysis demonstrates that winning the housing lottery at ages 20 to 25 increases the probability of having at least one child by approximately 32% and increases total expected children by approximately 33% relative to lottery losers. Critically, the fertility effect diminishes sharply with age at lottery entry. A 10-year delay in housing access is estimated to produce approximately 50% lower lifetime fertility, suggesting that the window during which housing access meaningfully influences fertility outcomes is concentrated in the late teens and early to mid-twenties, an age range during which housing access barriers have intensified most dramatically across OECD nations.

Detting and Kearney's (2014) analysis of US housing price variation finds that a \$10,000 increase in local house prices is associated with a 5% increase in fertility among existing homeowners and a 2.4% decrease in fertility among renters, for whom rising prices represent a barrier rather than a windfall. OECD housing dashboard data document that prime urban housing markets in the United Kingdom, Australia, Canada, New Zealand, and the United States require 10 to 20 years of full household income to reach purchase affordability thresholds, placing owner-occupation effectively beyond reach for young households during peak reproductive years without substantial parental wealth transfer.

Marriage rate data provides a parallel stream of causal evidence pointing to the same conclusion. US marriage rates have fallen approximately 60% since the 1970s, from 76.5 to approximately 30 marriages per 1,000 unmarried women (Herre et al. 2020). Japan's case is instructive: married couples average approximately 1.9 children, close to replacement, yet Japan's TFR has fallen to 1.2. The arithmetic gap is explained entirely by the collapse of marriage formation rates, not by any change in family size preferences among those who do form partnerships. Housing access and partnership formation are distinct mechanisms converging on the same outcome: stable household formation is the single most consistent prerequisite for fertility, and both routes to it are becoming harder for younger cohorts. The cultural forces driving partnership decline, including changing gender dynamics, digital technology, and shifting social norms, are examined in detail in Section 7.

5.6 The Sandwich Generation: The Hidden Double Burden

A structural dimension of fertility constraint that has received insufficient attention in demographic literature is the simultaneous care burden now borne by adults in their prime reproductive years. As of 2025, an estimated 63 million Americans, approximately 24% of the adult population, serve as family caregivers, a figure representing a 45% increase since 2015 (National Alliance for Caregiving and AARP 2025). Within this population,

approximately 29% are sandwich caregivers providing simultaneous support to both aging parents and children under 18.

The time intensity of eldercare is substantial, particularly for dementia care. Adults caring for a family member with dementia provide an average of 31 hours per week of unpaid care, compared to approximately 14 hours per week for adults caring for elders without dementia (National Alliance for Caregiving and AARP 2025). The gendered dimension of caregiving compounds fertility implications: approximately 75% of primary caretakers are female, meaning that the double burden of elder care and potential childcare falls disproportionately on women who are simultaneously navigating the peak years of career establishment.

A demographic timing collision amplifies the structural significance of this effect. The peak years of eldercare demand coincide precisely with the prime reproductive years of adults aged 30 to 45. The loss of grandparental childcare support represents an underappreciated driver of formal childcare cost escalation. In the mid-twentieth century, grandmothers who were not in the paid workforce provided substantial unpaid childcare to grandchildren. Today's grandparents are more likely to be employed into their late 60s and 70s, more likely to be geographically separated from adult children, and more likely to face their own health constraints that limit intensive childcare provision. The erosion of this historically substantial informal childcare resource represents a cost increase in child-rearing that is structurally invisible in standard analyses but fiscally and temporally significant for families navigating contemporary childcare markets.

6 Biological Factors: Is This a Medical Crisis?

Beyond economic and structural drivers, fertility decline has a biological dimension that social science analyses consistently underweight. Male reproductive health has experienced documented deterioration over the past half-century that cannot be explained by behaviour or preference. Levine et al.'s (2017) systematic review, pooling data from 185 studies across 43 countries covering 42,935 men between 1973 and 2011, documented a 52.4% decline in sperm concentration (the number of sperm per millilitre of semen) and a 59.3% decline in total sperm count among men from Western countries over the study period. A subsequent update extending the analysis through 2018 found that the rate of decline accelerated after 2000, with sperm counts falling at approximately 2.6% per year globally (Levine et al. 2022). The causes remain incompletely understood. The leading proposed contributors are endocrine-disrupting chemicals (substances that interfere with the body's hormone system, found in plastics, pesticide residues, and food packaging), sedentary lifestyles, heat exposure, and the hormonal disruption associated with rising obesity rates.

See Appendix Figure A7 for a visualisation of the sperm concentration decline from 1973 to 2018 and the post-2000 acceleration in the rate of decline, drawn from the Levine et al. meta-analyses.

The obesity epidemic represents an increasingly recognised biological pathway between metabolic and reproductive health. Approximately 73% of American adults are classified as overweight or obese, with 42% meeting the clinical threshold for obesity (Centers for

Disease Control and Prevention 2024). In men, excess body fat raises levels of an enzyme called aromatase, which converts testosterone into oestrogen. The result is lower testosterone, reduced sperm production, and lower libido. In women, excess body fat disrupts the hormonal signals that regulate ovulation (the monthly release of an egg), making conception harder. The US male infertility rate among men aged 15 to 49 stands at approximately 11.4%, with obesity-related hormonal disruption representing a growing contributor to this figure (Agarwal et al. 2021).

Advanced maternal age represents a further biological constraint whose demographic significance has grown as first-birth ages have risen. Female fertility declines substantially after age 35, with egg quality deteriorating through the late thirties and forties. In most high-income countries, the average age at first birth now falls between 27 and 30 years, with substantial numbers of women attempting first conception in their mid to late thirties. Women attempting conception at these ages face higher rates of subfecundity (reduced but not absent fertility), miscarriage, and chromosomal abnormality, even absent any underlying medical condition. The interaction between delayed childbearing and declining male fertility creates a compounding biological constraint for couples who delay family formation for economic or partnership reasons, a delay that the structural barriers documented elsewhere in this paper actively produce.

6.1 Infertility, Subfecundity, and Social Childlessness

A critical distinction is warranted between infertility (the clinical inability to conceive after 12 months of trying), subfecundity (reduced but not absent reproductive capacity), and social childlessness (no children despite biological capacity). The US average children per man aged 15 to 49 sits at approximately 0.9. Most childlessness in the data is social, not medical. Clinical infertility rates of 11 to 15% have not risen proportionally to childlessness rates, which have nearly doubled in some cohorts over two decades. Biological factors are real, worsening, and deserve clinical attention. But they explain a small fraction of the scale and pace of demographic decline. The primary drivers are structural, economic, and cultural. Improving reproductive medicine alone, through assisted conception (IVF), egg freezing (oocyte cryopreservation), or future technologies such as synthetic wombs (ectogenesis), will not reverse the trend. These technologies address the symptom for those who can afford them. They do not address the conditions that produced delayed or foregone parenthood in the first place.

The scale of the biological contribution relative to social childlessness is illustrated in Appendix Figure A7: the rate of sperm decline, though real and worsening, cannot arithmetically account for the near-doubling of childlessness rates documented in some cohorts.

7 Cultural Factors: Partnership Decline, Future Anxiety, and Neurochemical Displacement

7.1 Partnership Collapse

One of the most consequential yet underappreciated dimensions of fertility decline is the transformation of partnership formation patterns across Western and East Asian societies. Marriage rates in the United States have declined approximately 60% since the 1970s, falling from 76.5 marriages per 1,000 unmarried women aged 15 and older in 1970 to approximately 30.0 in recent years (Herre et al. 2020). While crude marriage rates are sensitive to compositional changes in the adult population, age-standardised measures confirm genuine declines in partnership formation rates, not merely population ageing artefacts. The proportion of adults who will never marry in their lifetimes has risen substantially across high-income countries, with permanent singlehood increasing in both the United States and Western Europe.

A structural paradox characterises this transformation: divorce rates, which rose sharply from the 1960s through the 1980s, have stabilised and in some measures declined since the early 1990s in the United States and several European countries. Marriages that do occur are increasingly selective of more compatible, stable, and economically secure partnerships. The result is a bimodal distribution: a stable minority who form durable, often relatively high-fertility partnerships, and a growing majority who experience delayed, interrupted, or absent partnership formation. This distributional shift, rather than a uniform decline in partnership quality, explains why aggregate marriage stability statistics can appear relatively favourable while rates of permanent singlehood rise simultaneously.

The gender asymmetry in partnership formation failures is particularly acute at the lower end of the socioeconomic distribution. Survey data indicate that approximately 28% of men under 30 reported zero sexual partners in 2023, up from approximately 10% in 2008 (Ueda et al. 2020, analysing General Social Survey data), a tripling over 15 years that is without historical precedent in survey records of sexual behaviour. The asymmetric pattern of sexual partner distributions, in which a minority of high-status men account for disproportionate numbers of partnerships while large numbers of men report zero romantic or sexual activity, reflects what sociologists have characterised as a hypercompetitive mating market produced by the intersection of female educational advancement, male economic marginalisation, and digital technology enabling both hypergamous sorting and substitution of parasocial romantic engagement for actual partnership.

The female side of this retreat from partnership is equally visible, if less statistically documented. A 2025 article in *British Vogue*, 'Is Having a Boyfriend Embarrassing Now?' by Chanté Joseph, captured a cultural shift that had been building across social media for several years: straight women increasingly viewing the public display of a boyfriend as socially uncool, with singlehood reframed as a mark of independence and self-sufficiency rather than a gap to be filled. The piece documented the rise of what commentators have termed heteropessimism, a generalised scepticism among women about the value of

heterosexual partnership, expressed in viral social media content, declining rates of women initiating contact on dating platforms, and a growing cultural script in which romantic relationships are framed as a net cost to women's autonomy rather than a net benefit. As one widely shared comment cited in the article put it: 'Boyfriends are out of style. They won't come back in until they start acting right.' This cultural mood does not exist in a vacuum. It is downstream of genuine structural grievances, the motherhood penalty, the unequal distribution of domestic labour, the capstone model of marriage that requires women to subordinate career momentum to partnership timing. But it also represents a feedback loop: as partnership becomes culturally coded as a threat to female autonomy, the social incentive to pursue it diminishes, independently of the structural conditions that produced the original scepticism (Joseph 2025).

The consequences for fertility are direct and arithmetically simple. Births within stable partnerships have remained relatively close to replacement for couples who form such partnerships. In low-partnership-support contexts like Japan, South Korea, and the United States, the decline of partnership formation rates closely tracks aggregate TFR decline, consistent with the extensive margin thesis. The marriage paradox, that marriages are more stable but fewer people are forming them, encapsulates the demographic reality: fertility has not collapsed among those with sufficient economic security to sustain partnerships. It has collapsed among those structurally excluded from stable partnership formation in the first place.

See Appendix Figure A9 for marriage rate time series across six countries from 1970 to 2023, and the share of adults never married by age 40-44, which quantifies the permanent singlehood trend underlying this dynamic.

7.2 Secularisation and Individualisation

Religion remains one of the few robust buffers against fertility decline. Data consistently show that highly religious communities, such as the Latter-day Saints in Utah or Orthodox populations in Israel, maintain fertility rates significantly above the replacement level, bucking the global trend (Götmark and Andersson 2020). This resilience is attributed not only to theological imperatives but to the community infrastructure that lowers the social cost of parenting. Conversely, the rapid secularisation of the West has eroded these norms. In a secular market society, the decision to reproduce becomes a purely utilitarian calculus of cost versus benefit, a calculation that, in the modern economy, almost strictly favours childlessness.

Ron Lesthaeghe's Second Demographic Transition framework situates this shift within a broader value transformation in post-industrial societies: the progressive prioritisation of individual self-realisation, expressive identity, and personal autonomy over institutional commitments including marriage and parenthood. Children have shifted from being a cornerstone of adult life, the foundational assumption around which other decisions were organised, to being a capstone: an optional achievement pursued only after career establishment, financial security, and personal development goals are met. In this framework, children become one option among many in a diversified life portfolio,

competing with career achievement, travel, consumption, and personal development. This is a genuine preference shift, not merely rationalised constraint, but one that operates downstream of structural conditions: economic security tends to correlate with higher fertility even within secular populations, suggesting that cultural individualisation and structural barriers interact rather than operate independently.

7.3 Concern About the Future of the World

A dimension of fertility decline rarely captured in economic models is collective pessimism about the future. Pew Research Center (2024) found that 32% of childless respondents cited concern about the state of the world as a major reason for not having children, a figure that cannot be explained by financial constraint or biological incapacity alone.

The generation now entering prime reproductive years has come of age through compounding uncertainty: the 2008 financial crisis, COVID-19, accelerating climate anxiety, and the emergence of artificial intelligence as both an economic and existential variable. For the first time, a cohort of prospective parents must genuinely reckon with the possibility that their children will enter a labour market substantially restructured by automation, not as a distant abstraction but as a present reality reshaping graduate employment now. Alongside this, rising geopolitical instability, the war in Ukraine, tensions in the Taiwan Strait, the fracturing of post-Cold War multilateral institutions, contributes to a diffuse sense that the stable, rules-based world in which family formation felt like a reasonable long-term investment may not hold.

This future-anxiety dimension cannot be addressed by financial transfers. It is a crisis of collective confidence, and one for which no obvious policy lever exists.

7.4 The Neurochemical Displacement Hypothesis

Dr. Alok Kanojia, a Harvard-trained psychiatrist and founder of HealthyGamer GG, has advanced a neurochemical hypothesis for declining partnership formation that extends the digital displacement argument from sociology into clinical neuroscience. Kanojia argues that the chronic, high-frequency stimulation produced by social media, operating through variable-reward notification structures analogous to those exploited by slot machines, progressively downregulates dopamine receptor sensitivity in the mesolimbic pathway. The brain, habituated to a high-frequency dopamine baseline, recalibrates its hedonic threshold upward. The consequence is that the quieter, slower dopamine signals associated with real-world romantic interest, the anticipation before a date, the salience of a specific person's attention, the reward of sustained pair-bond formation, register as insufficiently stimulating by comparison. Romantic attraction, in neurochemical terms, becomes harder to feel.

Kanojia further identifies a serotonin depletion mechanism: heavy social media use, by sustaining chronic low-grade social comparison and status anxiety, depletes serotonin availability. Serotonin is the neurochemical substrate of the obsessive, intrusive-thought quality of early-stage romantic love, the I cannot stop thinking about them phenomenology that has historically driven pair-bond formation. Its depletion may suppress the intensity of early romantic experience, making love feel less urgent, less consuming, and therefore less

motivating as a life-organising force. A 2024 ResearchGate study examining the effects of AI-driven romantic paradigms over a five-month period documented dopamine desensitisation and reduced emotional responsiveness to genuine social connection among participants, providing preliminary empirical support for the desensitisation mechanism (Volkow et al. 2012).

The population-level causal evidence for this hypothesis remains limited, and the directionality of observed correlations is not fully established. Lonely individuals may use social media more, rather than social media causing loneliness. Nevertheless, the mechanistic plausibility is well-grounded in established neuroscience, the timing correspondence between social media adoption curves and partnership formation decline is striking, and the hypothesis is consistent with the clinical observations of practitioners working with young adults experiencing relationship difficulties. It represents the least documented but potentially most structurally significant mechanism in this analysis: a process operating at the level of brain chemistry, independently of economic conditions, that makes the neurological experience of falling in love progressively harder for each successive cohort raised on high-frequency digital stimulation.

8 Market Responses: What the Free Market Has Tried

The paper's title asks how the free market will respond to global fertility decline. Having mapped the mechanisms of decline, it is possible to give a direct answer: the market has already responded to each mechanism, and in several cases, its responses have made the problem worse.

8.1 Biological Decline: The Fertility Economy

The market response to biological infertility has been swift, profitable, and structurally limited. A burgeoning fertility economy has emerged around assisted reproductive technologies: in-vitro fertilisation (IVF), oocyte cryopreservation (egg freezing), preimplantation genetic testing, and speculative research into partial ectogenesis (synthetic wombs). The global fertility services market is projected to exceed \$50 billion by 2030. These technologies address biological constraints for those who can afford them. IVF costs in the United States run \$15,000 to \$30,000 per cycle, with success rates that decline sharply with maternal age. The market has efficiently monetised biological anxiety, particularly among professional women in their thirties, without addressing the structural conditions that produced delayed childbearing in the first place. Egg freezing does not reverse the motherhood penalty. It defers it.

8.2 Partnership Failure: Dating Apps and Their Perverse Effects

The market response to partnership formation failure has been the dating app industry, now valued in the tens of billions globally. The empirical evidence on whether this response is net positive for fertility is, at best, mixed. Dating apps increase the volume of potential matches but appear to intensify hypergamous sorting, concentrating romantic attention on a small number of high-status individuals while leaving large numbers of men and women

unmatched. The swipe-based selection mechanism optimises for initial visual attraction rather than the compatibility dimensions that predict stable partnership. Longitudinal data on marriage rates show no recovery since the widespread adoption of dating apps in the 2010s. Marriage rates have continued to decline. The market has efficiently monetised the search for partnership without improving the rate at which partnerships form. There is a reasonable case that dating apps, by substituting the dopamine reward of matching and messaging for the harder work of sustained courtship, have contributed to the neurochemical displacement dynamic described above, making the market response to partnership failure part of the mechanism of partnership failure itself.

8.3 Neurochemical Displacement: AI Companions

The most striking market response to loneliness and partnership failure is the emergence of AI companion platforms, an industry valued at approximately \$208 billion. Products like Replika, Character.AI, and a growing cohort of competitors offer emotionally responsive, infinitely patient, algorithmically optimised conversational partners. The market has correctly identified that loneliness is a large, monetisable problem. What it has produced is a solution that addresses the symptom, the experience of loneliness, while potentially deepening the cause, the neurochemical displacement that makes real-world connection feel unrewarding by comparison. An AI companion is, by design, a higher-stimulation, lower-friction substitute for human relationship. Its existence does not increase the probability that its user will form a stable partnership and have children. It reduces it. The market is efficient at satisfying individual desires. It is not designed to solve collective long-term problems. The AI companion industry is the clearest example of the market accelerating demographic decline while profiting from it.

8.4 Economic Barriers: Pronatalist Transfers and Their Limits

Government and market responses to the economic barriers to fertility, pronatalist cash transfers, employer-funded fertility benefits, childcare subsidies, are the most direct attempt to address the problem at scale. As documented in the economics section, the evidence on their effectiveness is consistent: necessary, modestly effective, insufficient. Hungary's 6% of GDP programme produced TFR gains of 0.1 to 0.25. The OBBBA's \$1,000 Trump Account represents less than one week of US infant childcare costs. Nordic institutional models, which go furthest in reducing the motherhood penalty and providing universal childcare, achieve higher fertility than the OECD average, but not replacement. The market response to economic barriers is the most evidence-based and the most defensible. It is also the one that most clearly demonstrates that economics, while necessary, is not sufficient. If it were, Denmark would have solved this.

See Appendix Figure A8 for Hungary's TFR trajectory against its major policy milestones, and an OECD scatter of family policy spending against current TFR, which confirms that no country spending generously on pronatalist transfers has achieved replacement fertility.

8.5 The Furbaby Economy: Pets as a Partial Substitute

One market response to declining parenthood rates that rarely features in demographic literature deserves attention: the extraordinary growth of the pet industry. The US pet industry reached \$147 billion in revenue in 2023 and is projected to exceed \$152 billion in 2024 (American Pet Products Association 2024). Approximately 94 million US households now own at least one pet. Millennials and Gen Z lead multi-pet ownership, and 70% of childless millennial women describe their pets as substitutes for children. A 2025 survey found that 43% of Americans, with the figure significantly higher among younger cohorts, said they preferred pets to children.

The economics are instructive. The average annual cost of owning a dog in the United States runs to approximately \$2,759, compared to \$13,000 to \$18,000 per year for childcare alone. Pets offer companionship, purpose, and an object for caregiving instincts at a fraction of the financial and time cost of a child. They do not require school fees, university funding, or 18 years of sustained commitment. The market has, in effect, created a lower-cost, lower-risk substitute for one of the core emotional functions that children historically served. Whether this represents genuine preference satisfaction or a rationalisation of structural barriers to parenthood is precisely the preference endogeneity question raised in Section 4. What is clear is that the pet industry is growing fastest in exactly the demographic cohorts and geographies where fertility is falling fastest. The market has found a way to monetise the caregiving impulse without the caregiving commitment.

The aggregate picture is uncomfortable but clear. The free market has responded to fertility decline by building industries around each mechanism of that decline, and several of those industries are, structurally, fertility suppressants. The fertility economy, the dating app industry, the AI companion sector, and now the furbaby economy each extract economic value from the conditions that reduce fertility, without resolving them. The market is not going to solve this. It is going to adapt to it, profit from it, and in some respects accelerate it. The policy question is not whether the market will respond, but whether the response will be corrective or extractive. The evidence so far is predominantly the latter.

9 Is Declining Fertility Actually a Crisis?

Before concluding that declining birth rates represent an unambiguous catastrophe, it is worth sitting with a more uncomfortable question: what if they do not? A strand of serious academic and ecological thinking argues that a smaller human population is not inherently a crisis. Global resource consumption, carbon emissions, and biodiversity loss are all functions of population size. From a planetary perspective, fewer humans may represent a more sustainable equilibrium. The anti-natalist argument is not fringe. It is held by a significant minority of the generation now deciding whether to have children, and it appears explicitly in the Pew data: 32% of childless respondents cited concern about the state of the world as a major reason for not having children.

The economic case for alarm rests primarily on the dependency ratio: the ratio of working-age people to retirees. As this ratio deteriorates, the argument goes, pension systems

collapse and healthcare becomes unaffordable. But this argument assumes that human labour remains the primary driver of economic output. That assumption is increasingly uncertain. If artificial intelligence and automation can perform the productive work of three humans for every one, then the dependency ratio as currently calculated may become the wrong metric entirely. A smaller, healthier, better-educated population supported by highly productive AI systems may prove more economically viable than a larger population with a favourable age structure but lower productivity. The decoupling of GDP from headcount, already visible in productivity data, may ultimately render the demographic dividend argument obsolete.

There is a further variable that demographic models have not yet incorporated in any serious way: the possibility of meaningful life extension. Research into the biology of ageing, from senolytics (drugs that clear damaged cells) to partial cellular reprogramming, is advancing rapidly. If the average healthy lifespan extends materially over the next two to three decades, the entire dependency ratio calculus changes. A 75-year-old who is cognitively and physically capable of working is not a dependent. The boundary between working age and retirement age is a social and political construct, not a biological fixed point. Longevity science does not solve the fertility question, but it substantially changes the urgency of the answer.

10 Conclusion

Five interconnected mechanisms drive fertility decline in advanced economies, and no single one is sufficient to explain the scale of what is being observed. First, partnership formation failure: fewer adults, particularly economically marginalised men, are entering the stable household arrangements that historically preceded childbearing. Second, structural economic barriers: the child cost explosion, the motherhood earnings penalty, housing unaffordability concentrated precisely in the early to mid-twenties window when housing access most influences lifetime fertility, and the invisible double burden of sandwich caregiving. Third, biological degradation: sperm quality decline, obesity-driven hormonal disruption, and the biological consequences of delayed parenthood, real, worsening, but insufficient to explain the bulk of demographic change. Fourth, cultural displacement: secularisation, individualisation, and a spreading pessimism about the kind of future available to the next generation. Fifth, neurochemical displacement: the progressive recalibration of the dopamine reward system by digital technology, making the quieter signals of romantic attraction and pair-bond formation harder to register against a chronically over-stimulated baseline.

One of the paper's most counterintuitive findings warrants explicit restatement: the education-fertility U-curve. The women least likely to have children in contemporary advanced economies are not the most or least educated. They are the middle tier: college-educated, earning between £50,000 and £100,000, maximally squeezed between opportunity costs too high to ignore and childcare costs too large to absorb. This is not a failure of ambition. It is a structural failure of the economic arrangements governing family formation in middle-income households.

Before concluding that declining birth rates represent an unambiguous catastrophe, it is worth sitting with a more uncomfortable question: what if they do not? A strand of serious academic and ecological thinking argues that a smaller human population is not inherently a crisis. Global resource consumption, carbon emissions, and biodiversity loss are all functions of population size. From a planetary perspective, fewer humans may represent a more sustainable equilibrium. The anti-natalist argument is not fringe. It is held by a significant minority of the generation now deciding whether to have children, and it appears explicitly in the Pew data: 32% of childless respondents cited concern about the state of the world as a major reason for not having children.

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What is certain is that the United Nations has consistently overestimated future birth rates in its long-run projections. We are living in increasingly uncertain times, with technological, geopolitical, and cultural variables shifting faster than any demographic model can reliably track. Watching how this trend continues will be increasingly important for governments, economists, and anyone trying to understand what the world of 2050 will actually look like.

The evidence assembled in this analysis suggests that fertility decline will not be reversed by financial transfers alone, nor by medical technology, nor by social media campaigns. It will require restoring the conditions under which young adults anticipate a future worth reproducing into.

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APPENDIX OF CHARTS AND FIGURES

How Will the Free Market Respond to Global Birth Rate Decline?

Annabelle Body · March 2026

This appendix presents nine visual figures supporting the analysis in the main paper. Figures A1–A6 are sourced from Our World in Data, Statista (World Bank / UN DESA), UNFPA, and the US Census Bureau, all reproduced under their respective open-access or educational-use terms. Figures A7–A9 are published charts from Statista (A7, A8) and Our World in Data (A9), reproduced with full source attribution.

Contents

Figure	Title	Section in paper
A1	Global fertility rate by country, 2023	Section 3.1
A2	Global birth rate, death rate & natural population change, 1950–2100	Section 3.1
A3	Age of mothers at childbirth by cohort — tempo effect	Section 4.1
A4	World population by age group, 1950–2100 projections	Section 3.1
A5	US population pyramids, 2000 / 2010 / 2020	Section 3.1
A6	Reasons for not having children — 14-country survey, 2024	Sections 4.1, 5, 7.3
A7	Levine et al. (2022), via Statista (statista.com/chart/10439/)	Published infographic
A8	Hungarian Central Statistical Office, via Statista (statista.com/statistics/1127773/)	Published chart
A9	OECD (2025) and other sources, via Our World in Data (ourworldindata.org)	Published chart

Section 1: The Scale and Geography of Global Fertility Decline

The following two charts document the depth and reach of the fertility transition described in Section 3. The global TFR has more than halved since the 1960s and the decline now encompasses countries housing two-thirds of humanity. The second chart reveals the critical longer-run implication: birth and death rates are projected to converge and cross before 2100, at which point natural population decline becomes structural.

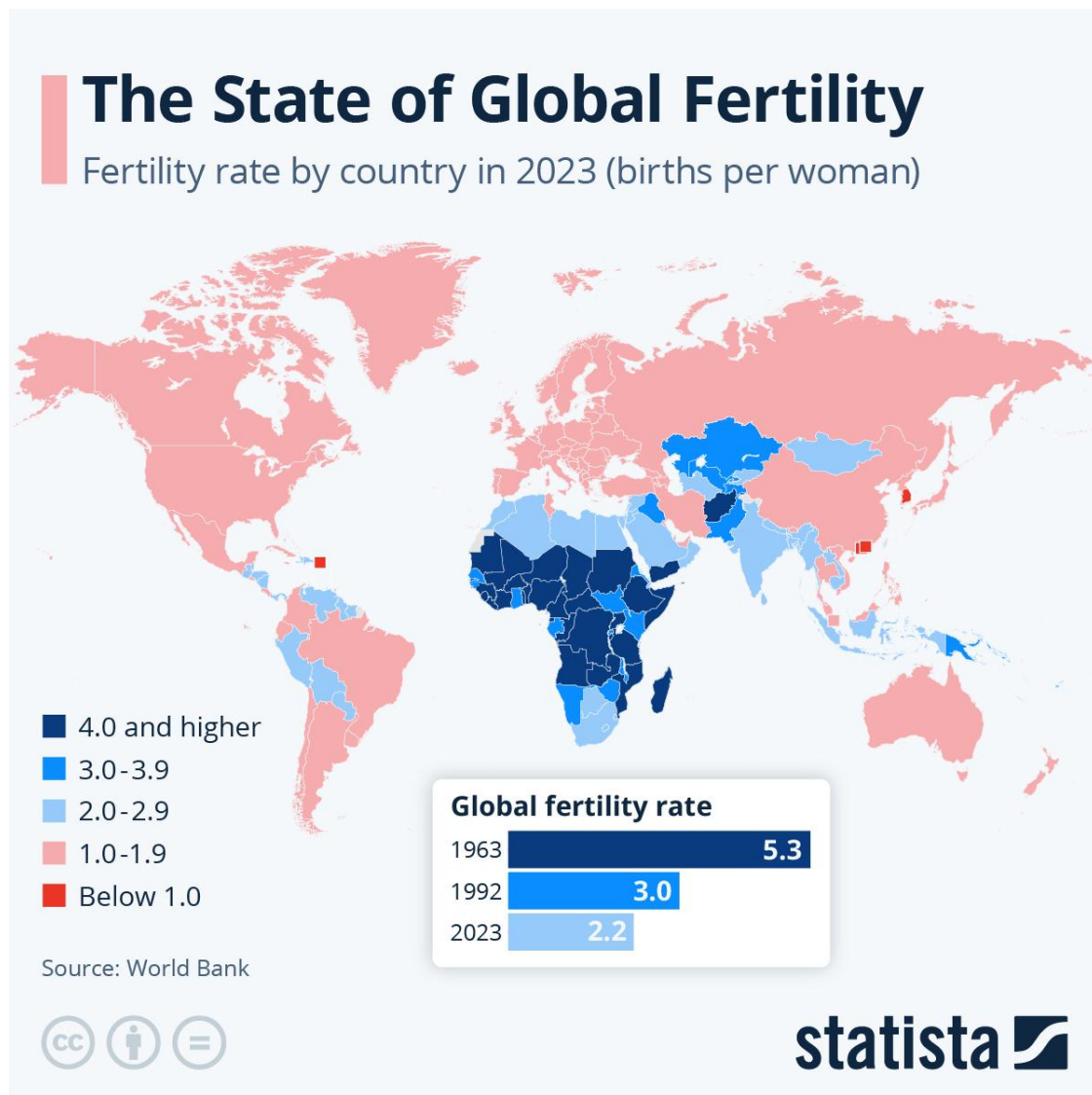


Figure A1: Global fertility rate by country, 2023 (births per woman)

Source: World Bank (2024), via Statista. Global average fell from 5.3 (1963) to 2.2 (2023). Red dots = countries below TFR 1.0, including South Korea (0.72) and Taiwan (0.87). The almost entirely pink (1.0–1.9) colouring of Europe, East Asia and the Americas shows the full geographic extent of below-replacement fertility.

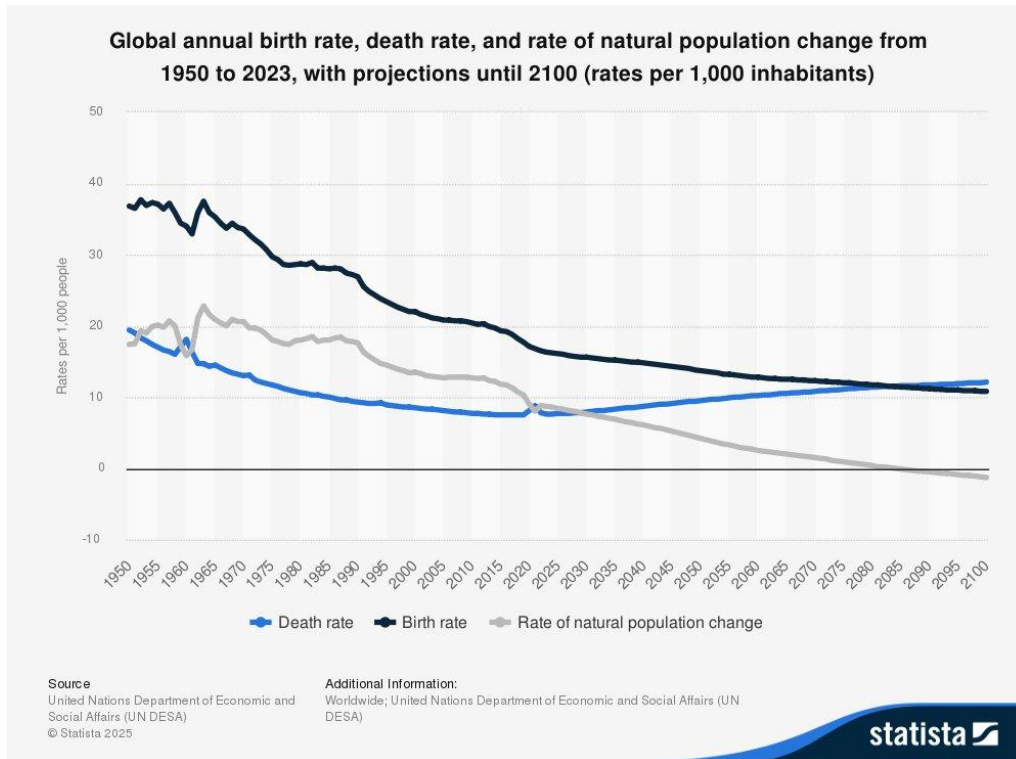


Figure A2: Global annual birth rate, death rate, and rate of natural population change, 1950–2100 (per 1,000 inhabitants)

Source: UN DESA, via Statista (2025). Birth and death rates converge circa 2023; the rate of natural population change (grey) turns negative in projections before 2100. This confirms the paper's argument that global population growth is sustained only by demographic momentum, not by current fertility rates.

Section 2: The Tempo Effect — Delayed Childbearing, Not Just Smaller Families

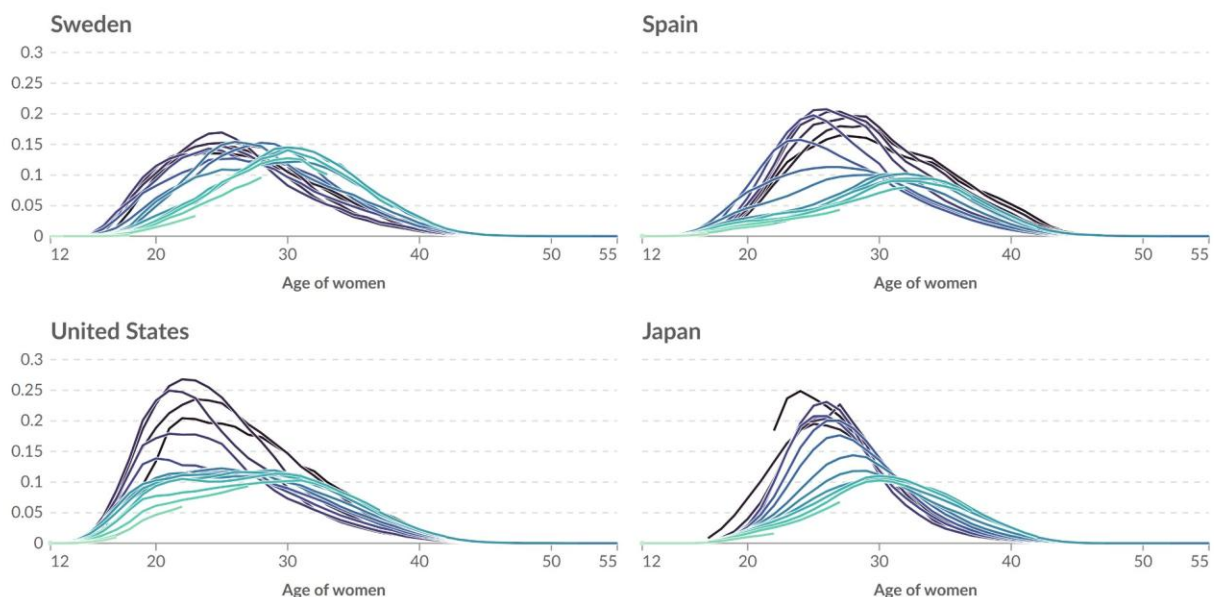
Section 4.1 introduces the extensive margin thesis: fertility decline is concentrated in fewer adults entering parenthood, not in smaller families among those who do. The Our World in Data cohort chart below provides direct visual evidence of the tempo shift — each successive cohort of mothers has been older at first and subsequent births, with youngest cohorts' reproductive curves barely visible.

Age of mothers at childbirth by mother's birth year

Our World
in Data

Each line denotes women born in a particular year, and shows their timing of childbearing. The vertical axis shows the age-specific fertility rate¹ per woman.

2010 2005 2000 1995 1990 1985 1980 1975 1970 1965 1960 1955 1950
1945 1940 1935 1930 1925



Data source: Human Fertility Database (2025)

OurWorldinData.org/fertility-rate | CC BY

1. Age-specific fertility rate The fertility rate (number of births per woman) in a specific age group.

For example, if 80 babies are born to 1,000 women aged 25–29 in a year, the age-specific fertility rate at ages 25–29 would be 80 per 1,000 women, or 0.08 per woman.

This can be calculated for women born in the same year (where it is known as the cohort age-specific fertility rate) or for a given year (where it is the period age-specific fertility rate).

Figure A3: Age of mothers at childbirth by mother's birth year — Sweden, Spain, United States, Japan

Source: Human Fertility Database (2025), via Our World in Data (ourworldindata.org/fertility-rate, CC BY). Each curve represents women born in a given year; the y-axis is the age-specific fertility rate per woman. The rightward shift of the peak (later first births) and the compression of total curve area (fewer births overall) is visible in all four countries. The US shows the most dramatic collapse in fertility at ages 15–25.

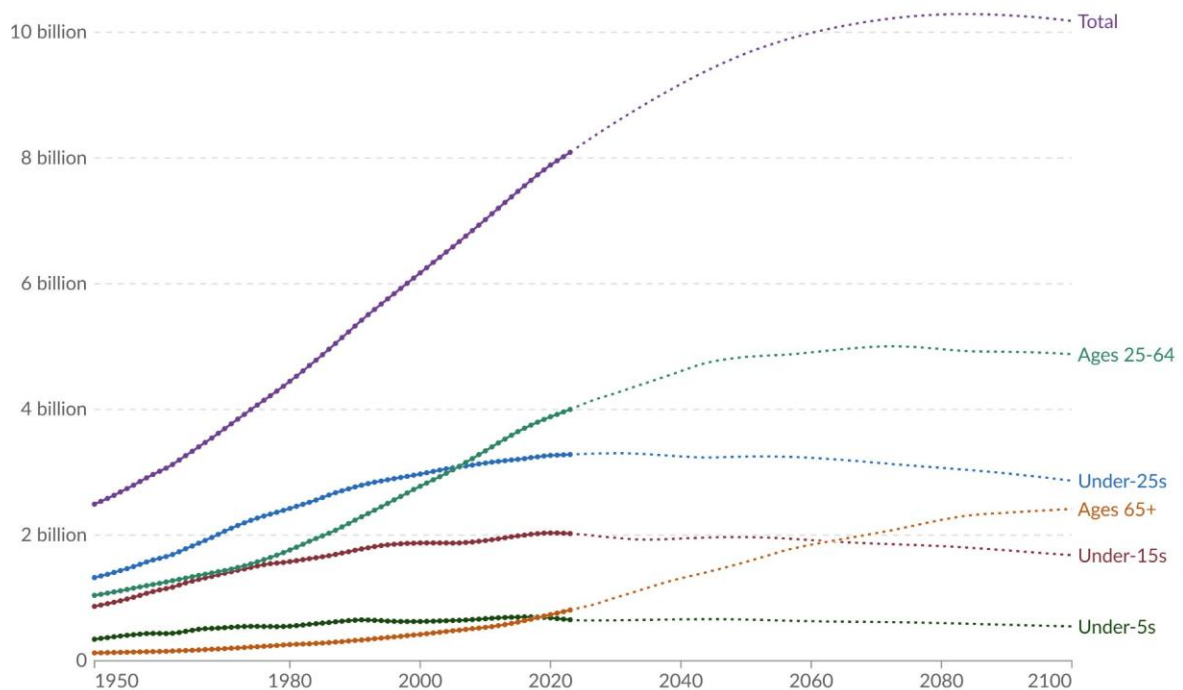
Section 3: From Pyramids to Obelisks — Structural Population Consequences

The paper's dependency ratio analysis (Section 3.1) and its discussion of pension system stress (Section 5.6) depend on understanding how age structures are inverting. The two charts below show the global picture of shifting age-group populations to 2100, and the US pyramid transformation as a concrete national illustration of the broader pattern McKinsey Global Institute terms the 'pyramid to obelisk' transition.

Population by age group, World

Our World
in Data

Historic estimates from 1950 to 2023, and projected to 2100 based on the UN medium scenario¹.



Data source: UN, World Population Prospects (2024)

OurWorldinData.org/population-growth | CC BY

1. United Nations projection scenarios The United Nations' World Population Prospects include a range of projected scenarios for population change. These scenarios rely on different assumptions for fertility, mortality, and migration patterns to project different demographic futures.

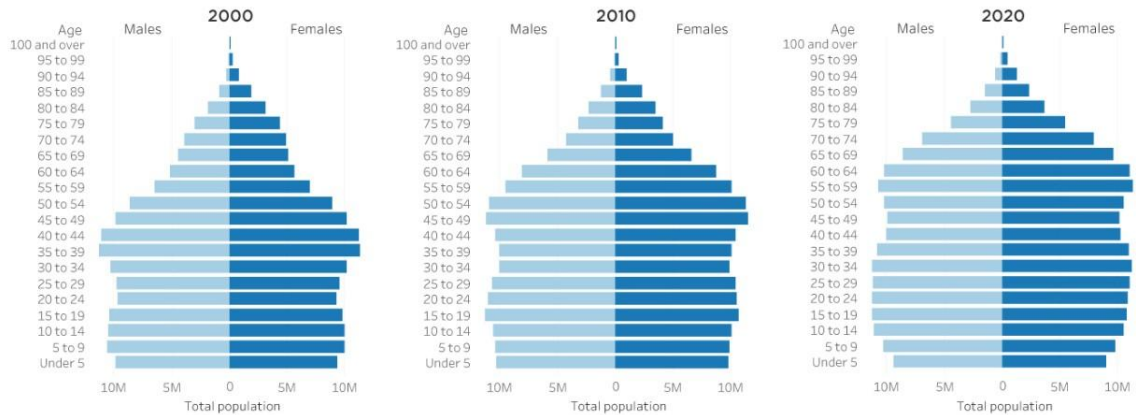
[Read more on population.un.org](https://population.un.org/)

Figure A4: World population by age group, 1950–2023 with projections to 2100 (UN medium scenario)

Source: UN World Population Prospects (2024), via Our World in Data (ourworldindata.org/population-growth, CC BY). Under-25s (blue) are projected to decline in absolute terms after approximately 2060; Ages 65+ (orange) rise steeply and continuously. The crossover of 'under-5s' with '65+' is projected around 2065 — directly supporting the paper's dependency ratio trajectory in Section 3.1.

Figure 2.

Population Pyramids for the United States: 2000, 2010 and 2020
(In millions)



Source: U.S. Census Bureau, Census 2000 Summary File 1 (SF1), 2010 Census Summary File 1 (SF1) and 2020 Census Demographic and Housing Characteristics File (DHC).

Figure A5: Population pyramids for the United States: 2000, 2010, and 2020

Source: US Census Bureau (2000 SF1; 2010 SF1; 2020 DHC). The classic triangular pyramid of 2000 gives way to a near-rectangular obelisk by 2020, with bulging mid-age cohorts and a narrowing base. The Baby Boomer cohort (born 1946–1964) is visible moving up through the pyramid across the three snapshots, and will be fully in the 65+ bracket by 2030.

Section 4: Reasons for Not Having Children — Cross-National Survey Evidence

Section 4.1 notes the difficulty of decomposing voluntary from involuntary childlessness, and Sections 5, 7.1, and 7.3 examine the economic, partnership, and cultural mechanisms in turn. The UNFPA chart below provides direct survey evidence of the multi-causal structure of childlessness across fourteen countries, supporting the paper's multi-mechanism analytical framework.

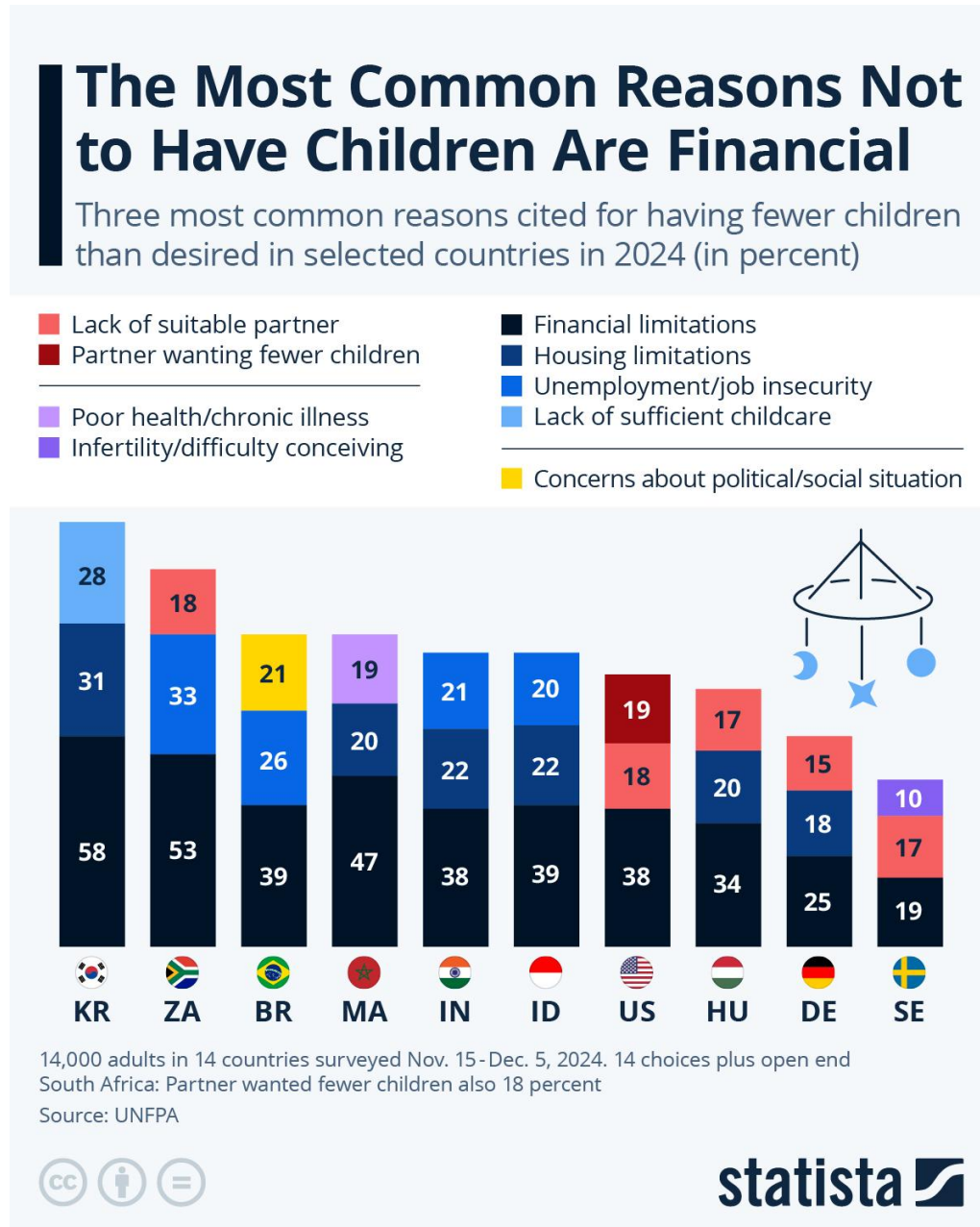


Figure A6: Three most common reasons cited for having fewer children than desired (% of respondents), selected countries, 2024

Source: UNFPA survey of 14,000 adults across 14 countries, Nov–Dec 2024, via Statista. Financial limitations (black) dominate in all markets. Note that 'concerns about political/social situation' (yellow, Brazil 21%) supports Section 7.3's future-anxiety argument; 'infertility/difficulty conceiving' (purple, Morocco 19%) supports Section 6; 'lack of suitable partner' (pink, South Africa 18%) supports Section 7.1.

Section 5: Biological Decline — Sperm Count and Reproductive Health

Section 6 establishes that biological factors are real, worsening, and deserve clinical attention, while cautioning that they cannot alone explain the scale of demographic decline. The chart below visualises the Levine et al. (2017, 2022) meta-analysis findings — the most comprehensive evidence base on male reproductive health trends — and shows both the magnitude of the decline and its post-2000 acceleration.

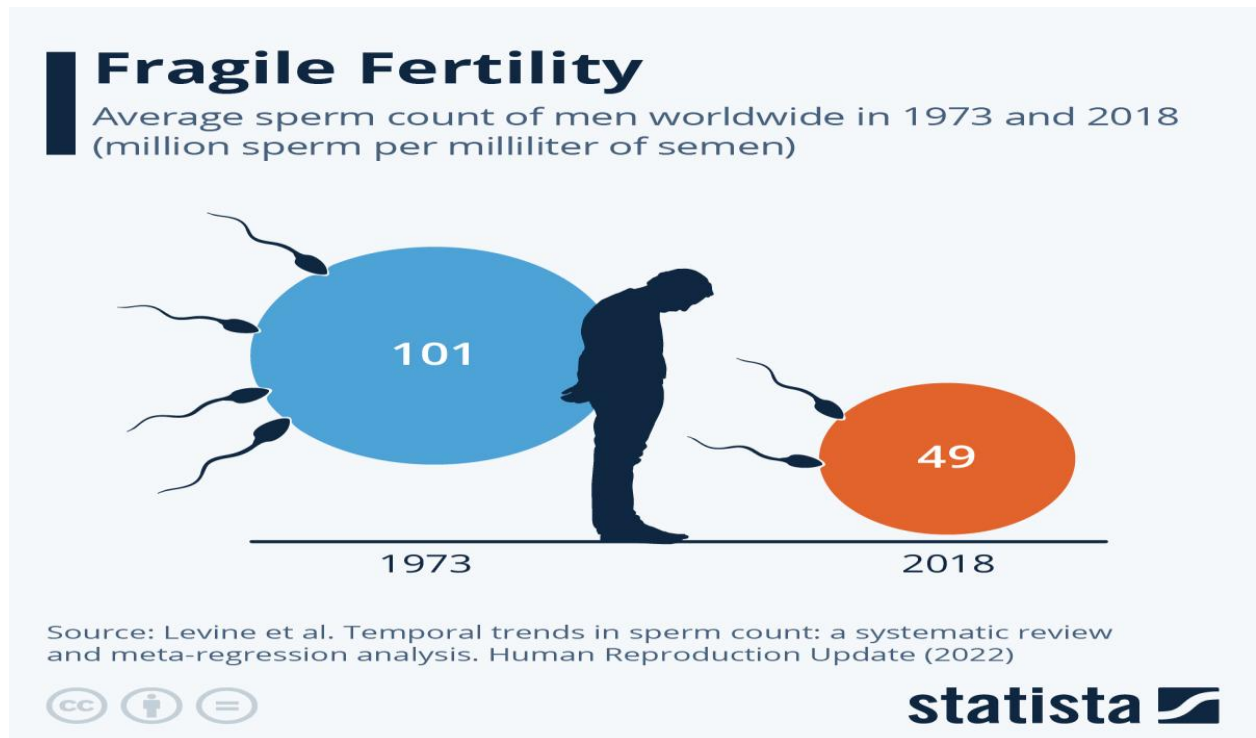


Figure A7: Average Global Sperm Count, 1973 vs. 2018 (million sperm per millilitre of semen)

Source: Levine et al. (2022). Temporal trends in sperm count: a systematic review and meta-regression analysis. Human Reproduction Update 29(2):157–176. Via Statista ([statista.com/chart/10439/](https://www.statista.com/chart/10439/)). © Statista 2026.

Section 6: Pronatalist Policy — Necessary but Not Sufficient

Section 8.4 argues that economic transfers are the most evidence-based market response to fertility decline, yet no country has achieved replacement fertility through policy alone. The two panels below track Hungary's TFR against its major policy interventions (the most ambitious pronatalist programme in the world at ~6% of GDP), and place all OECD countries on a spending-vs-TFR scatter to test whether higher spending reliably produces higher fertility. The answer is a modest positive correlation — overwhelmed by other structural forces.

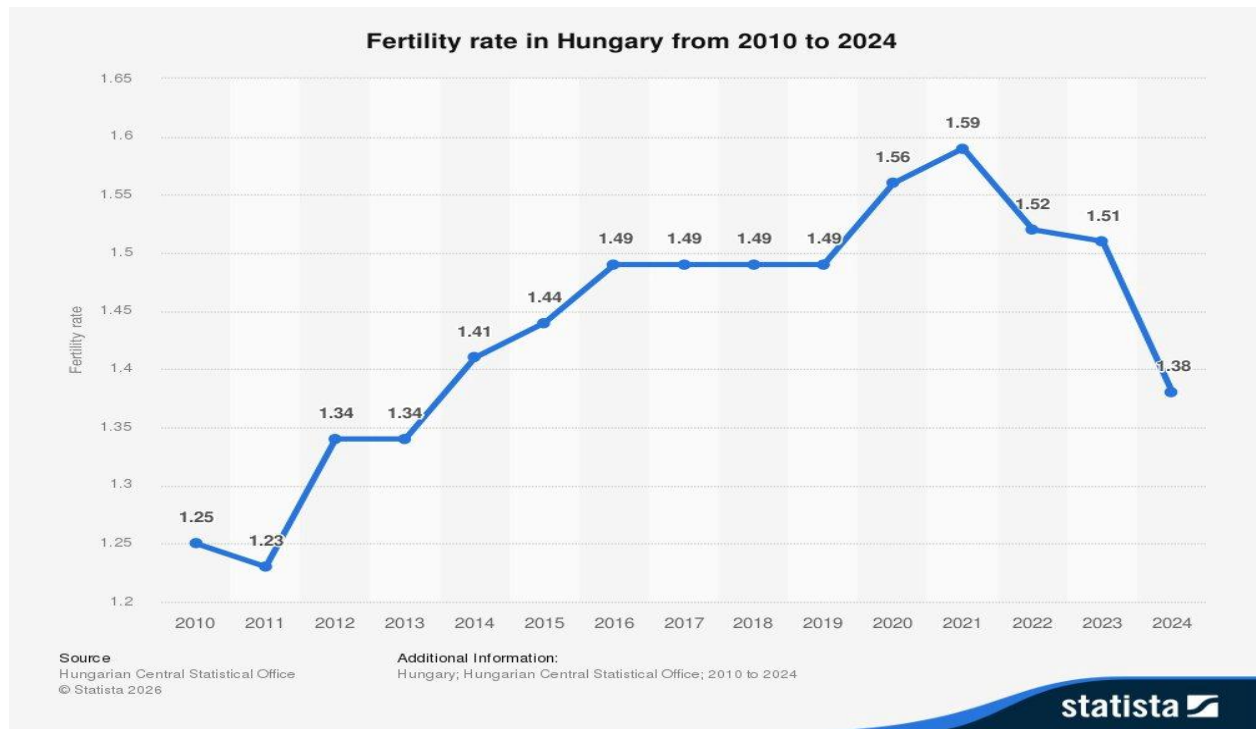


Figure A8: Fertility Rate in Hungary, 2010–2024 (total births per woman)

Source: Hungarian Central Statistical Office (2024), via Statista

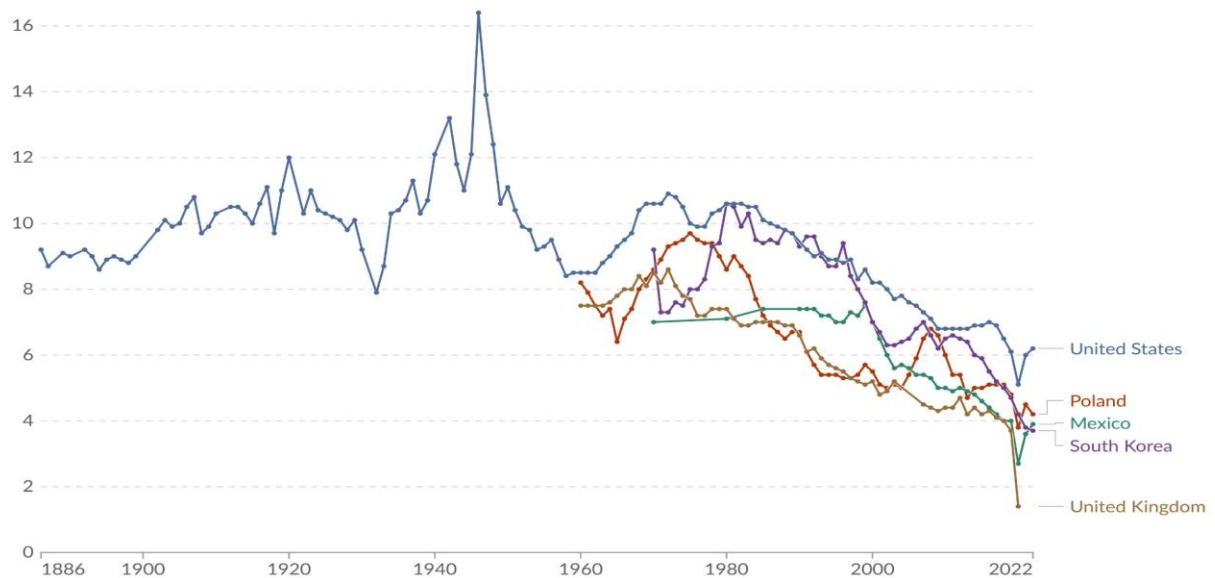
(statista.com/statistics/1127773/). © Statista 2026. TFR peaked at 1.59 in 2021 before falling to 1.38 by 2024, despite Hungary spending ~6% of GDP on pronatalist policy — confirming modest tempo gains but no quantum reversal of decline.

Section 7: Partnership Formation Collapse

Section 7.1 identifies partnership collapse as one of the most consequential yet underappreciated dimensions of fertility decline, citing the ~60% fall in US marriage rates since 1970 and Japan's Japan paradox (married couples average 1.9 children, but TFR is 1.2 because marriage formation has collapsed). The two panels below provide the international time-series evidence and quantify the share of adults who will never marry — the 'permanent singlehood' trend that is the proximate driver of the extensive margin decline.

Marriage rates

Number of marriages during a given year per 1,000 people.



Data source: OECD (2025) and other sources

OurWorldinData.org/marriages-and-divorces | CC BY

Figure A9: Marriage Rates by Country, 1886–2022 (marriages per 1,000 people)

Source: OECD (2025); OECD (2024); US Census Bureau (1949); Centers for Disease Control and Prevention (2020) — with minor processing by Our World in Data (ourworldindata.org/marriages-and-divorces). CC BY. South Korea fell from ~10 marriages/1,000 in 1970 to ~3.9 by 2022; the UK collapsed to ~1.5 by 2022.

Section 4.2 identifies male childlessness as one of the most consequential blind spots in demographic research and policy. Male childlessness consistently exceeds female childlessness in every country where data exists, yet it is almost entirely absent from policy discourse. The Norway income gradient (right panel) is among the most striking findings in the demographic literature: at the lowest income deciles, nearly three-quarters of men remain childless — a 61 percentage-point gap versus top earners, widening over three decades.

Notes on Sources, Numbering, and Merging

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